Example Analyses using countmods

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1 Introduction

In this quarto document, I will analyze a variety of data sets with emphasis on showcasing the `countmods` package.

AUTHORS NOTE: The Liklihood Ratio Test has a bug where p-values are calculated incorrectly, it has been fixed in the latest version of the package.

```
set.seed(1717)
library(tidyverse)
library(countmods)
```

2 Armadillos Dataset

Question: How does Age impact armadillos hunting ability?

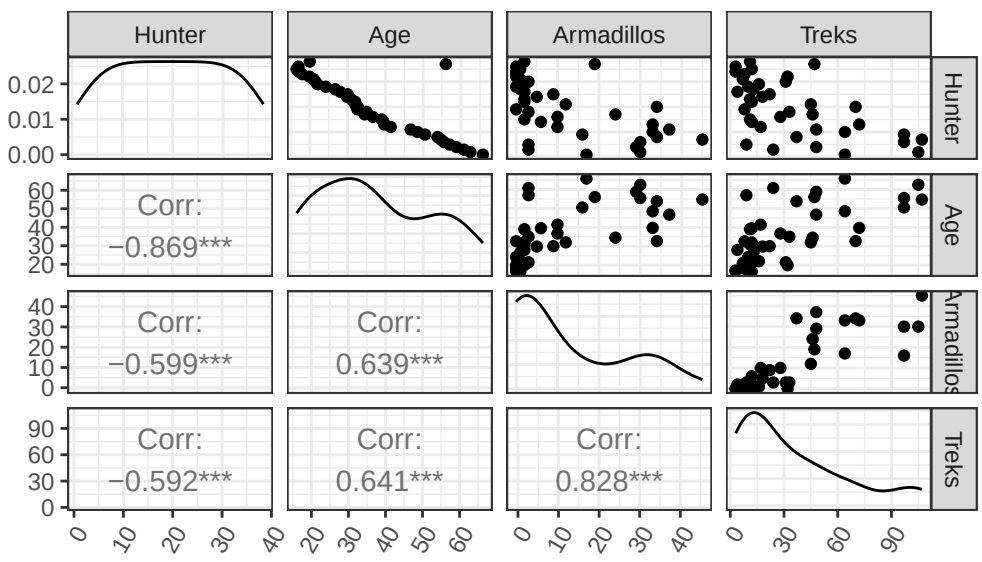
I will start with the Armadillos dataset (McMillan 2001). Below, I load in the dataset for Armadillos:

```
arma.dat <- read_csv("McMillanAcheArmadillo.csv")
```

We can begin by summarizing the data using `dat_plot()`.

```
dat_plot(arma.dat)
```

Figure 1: Scatter plot matrix of Armadillos data



It appears as if Age has a dimminishing relationship with Armadillos.

Now, we can fit the models using `glm_pois()` and `glm_negb()`. Since the amount of armadillos captured may depends on the number of treks, we will offset the models by `Treks`

```
mod1 <- glm_pois(data=arma.dat, Armadillos~Age+I(Age^2), offsetparm = "Treks")
mod2 <- glm_negb(data=arma.dat, Armadillos~Age+I(Age^2), offsetparm = "Treks")
```

Using `resid_diag()`, we can perform diagnostics on the model fits using randomized quantile residuals.

```
resid_diag(mod1)
resid_diag(mod2)
```

Figure 2: Resiudal Diagnostics of Poisson and Negative Binomial Models

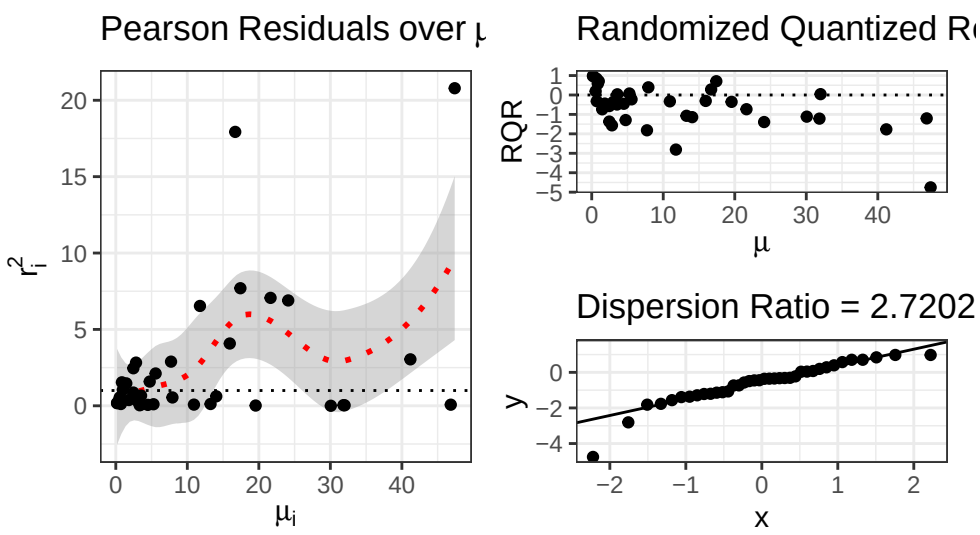
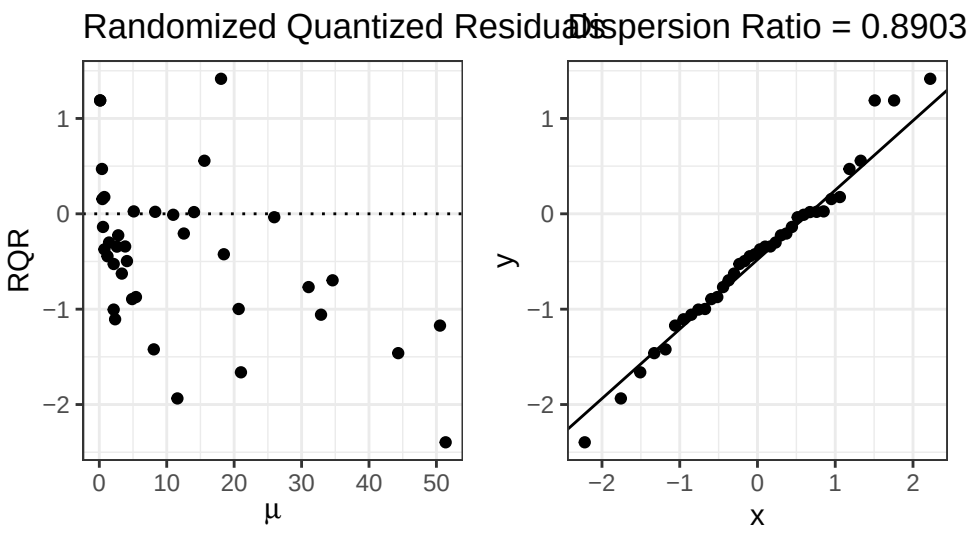


Figure 3: Resiudal Diagnostics of Poisson and Negative Binomial Models



As seen by the plots, we appear to have some over-dispersion in the data as seen in the randomized quantile residulas in the poisson model. The negative binomial model has a lower dispersion ration than the poisson model, so it seems to be a better fit.

To investigate further, we can use `mod_compare()` to compare model fits.

```
mod_compare(mod1, mod2)
```

```
---------------------------
Model 1:
AIC: 236.09
BIC: 241
R^2 (McFadden): 0.22
logLiki: -115.05
---------------------------
Model 2:
AIC: 201.22
BIC: 207.77
R^2 (McFadden): 0.13
logLiki: -96.61
---------------------------
Liklihood Ratio Test:
 ²: 6.443538e-10
p-val: 1.256281e-09
---------------------------

$mod1
[1]  236.09  241.00    0.22 -115.05

$mod2
[1] 201.22 207.77   0.13 -96.61
```

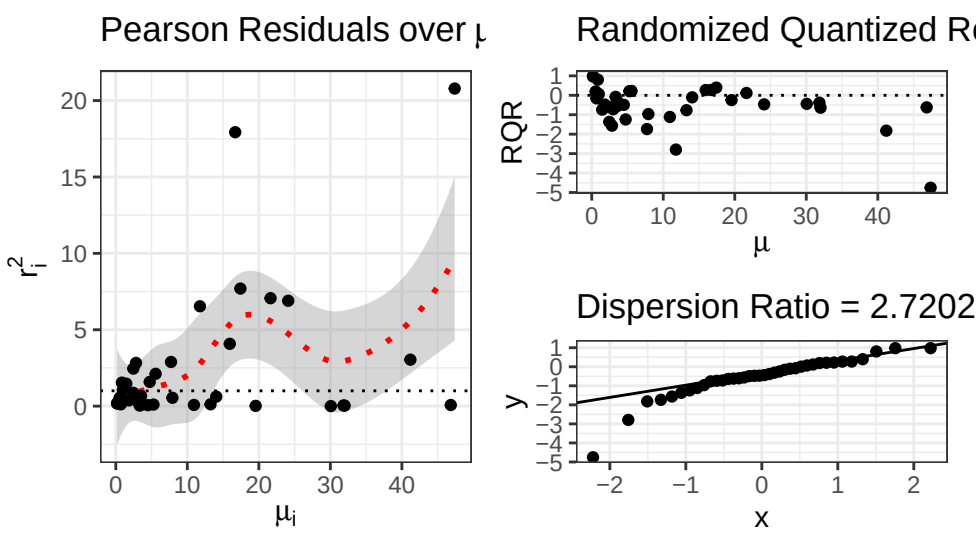
Intrestingly enough, the poisson model seems to be a better fit than the negative binomial model by most metrics in `mod_compare()` (NOTE: ignore the liklihood ratio test, the models are not nested, so the statistic isn't valid). Lets try a quasi poisson model aswell!

```
mod3 <- glm_pois(data=arma.dat,
                 Armadillos~Age+I(Age^2),
                 offsetparm = "Treks",
                 quasi = T)
```

and perform diagnostics

```
resid_diag(mod3)
```

Figure 4: Residual Diagnostics of Quasi-Poisson Model




```
mod_compare(mod1, mod1)
```

```
---------------------------
Model 1:
AIC: 236.09
BIC: 241
R^2 (McFadden): 0.22
logLiki: -115.05
---------------------------
Model 2:
AIC: 236.09
BIC: 241
R^2 (McFadden): 0.22
logLiki: -115.05
---------------------------
Liklihood Ratio Test:
 ²: Inf
p-val: 1
---------------------------

$mod1
[1]  236.09  241.00    0.22 -115.05

$mod2
[1]  236.09  241.00    0.22 -115.05
```

not much of a difference. For all intents and purposes, let use both inital models.

Now, we can get a summary of the data, lets use `interpert()` to do this!

```
interpret(mod1)
```

```
Call:
glm_pois(data = arma.dat, formula = Armadillos ~ Age + I(Age^2),
    offsetparm = "Treks")

Residuals:
      Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
-31.384980  -1.465133  -0.260594  -0.000009   1.113356  17.302306

Count Coefficients:
              Estimate   Std.error Crit.stat      P.value
(Intercept) -5.9064203 0.721342440 -8.188095 2.220446e-16
Age         24.8298075 0.033152817  6.689660 2.236900e-11
I(Age^2)    -0.2353829 0.000366315 -6.433271 1.248865e-10
```

```
interpret(mod2)
```

```
Call:
```

```
glm_negb(data = arma.dat, formula = Armadillos ~ Age + I(Age^2),
    offsetparm = "Treks")

Residuals:
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
-35.3678  -1.5358  -0.4209  -0.4091   1.1556  15.9465

Count Coefficients:
              Estimate    Std.error Crit.stat      P.value
(Intercept) -6.8024228 1.1096216526 -6.130398 8.765928e-10
Age         29.6964567 0.0535057192  4.859790 1.175100e-06
I(Age^2)    -0.2727226 0.0006078467 -4.492829 7.028307e-06
```

As we can see, we have a 24% increase in armadillos as age increases a single unit with the poisson model and a 29 increase in armadillos as age increase a single unit with the negative binomial.

Any further analyses should be done using `emmeans` or other supplementary packages.

3 Bio Chemists Dataset

Question: How do some factors (such as kids, phd prestige, or martial status) impact a bio chemist's ability to publish articles?

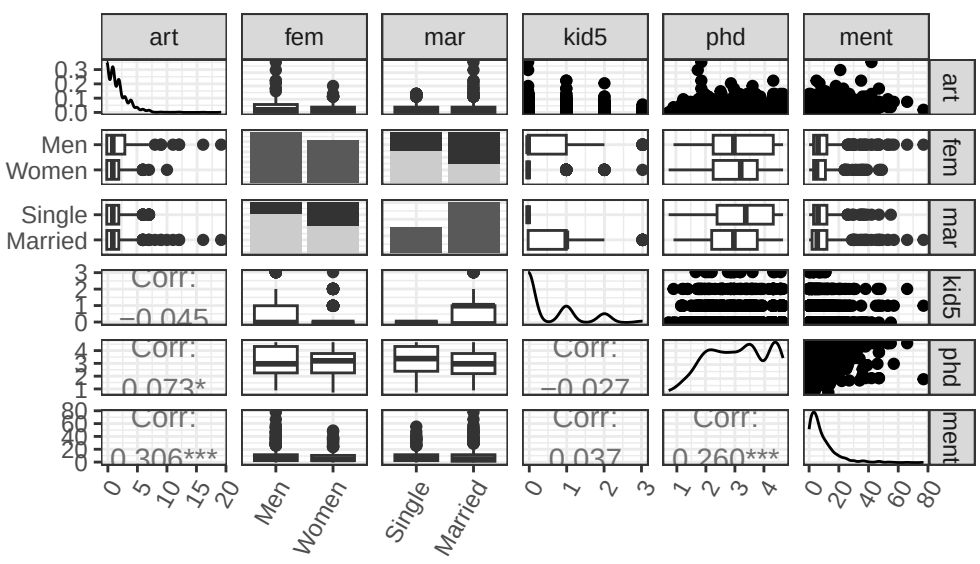
Now, we will use the `bioChemists` (Long 1990) dataset from the `pscl` (Jackman 2008) package, as this dataset is known for being zero-inflated. Below, I load in the dataset:

```r
utils::data(bioChemists, package = "pscl")
```

and we can do a quick plot summary with it:

```r
dat_plot(bioChemists)
```

Figure 5: Scatterplot Matrix of bioChemists Data



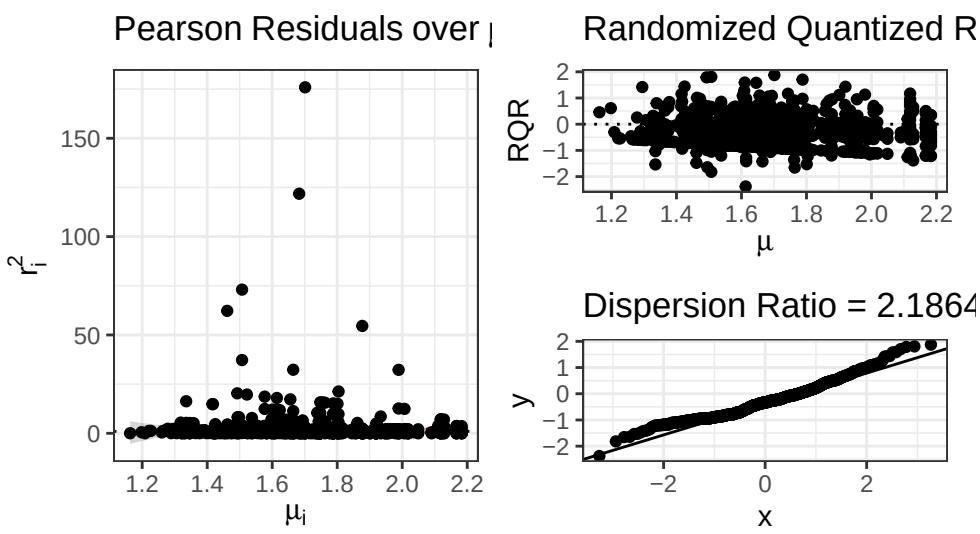
Interesting! Now lets fit some models using `glm_pois()`! This dataset primarily focuses on the relations between the amount of articles produced with by bio chemists and factors in the data set, such as having kids under 5 (`kid5`), phd quality (`phd`), and martial status (`mar`).

```r
mod1 <- glm_pois(data=bioChemists, art~kid5+phd+mar)
```

Now, we can do some diagnostics with `resid_diag()`.

```r
resid_diag(mod1)
```

Figure 6: Residual Diagnostics of Poisson Model for biochemists

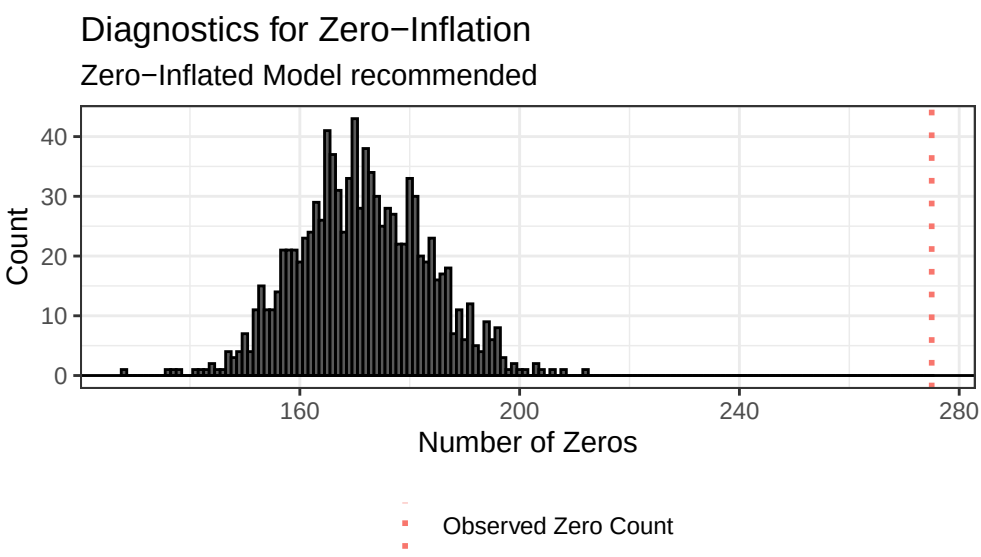


Some bad signs with the way those residuals are acting!

We know that the data is zero inflated, so lets run `zeroinf_test()` to check!

```r
zeroinf_test(mod1)
```

Figure 7: Zero-Inflation Test bioChemists Data



As we can see the data is zero-inflated, Yikes!

Luckily, we have options. We can use `glm_negb_zero()` to handle the overdispersion. Below, I fit a zero inflated model using `glm_negb_zero()`: We know that the data is zero inflated, so lets run `zeroinf_test()` to check! For the logistic part, we will use `kid5`.

```
mod.zero <- glm_pois_zero(data=bioChemists, art~kid5+phd+factor(mar), art~kid5)
mod.zero.negb <- glm_negb_zero(data=bioChemists, art~kid5+phd+factor(mar), art~kid5)
```

Now the zero should be handled, lets do some diagnostics with `resid_diag()`!

```
resid_diag(mod.zero)
resid_diag(mod.zero.negb)
```

Figure 8: Residual Diagnostics of Zero-Inflated Models

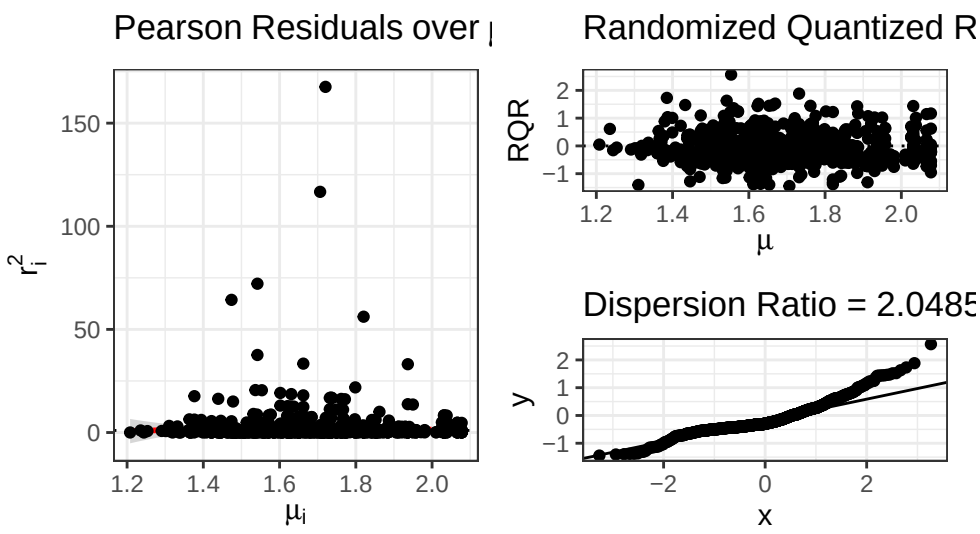
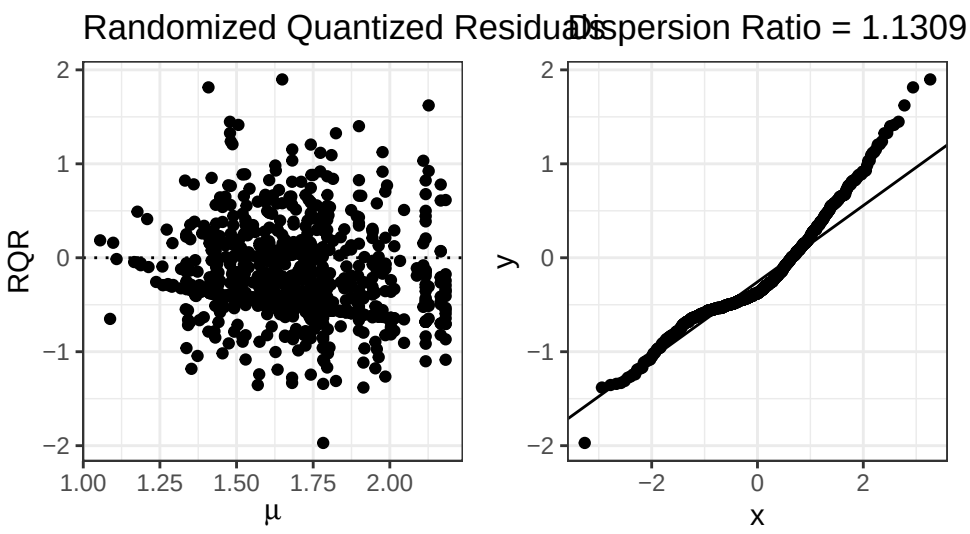


Figure 9: Residual Diagnostics of Zero-Inflated Models



Since the zero-inflated negative binomial has less dispersion, we will use this model! We can now interpret the model with `interpret()`

```
interpret(mod.zero.negb)
```

```
Call:
glm_negb_zero(data = bioChemists, formula.negb = art ~ kid5 +
    phd + factor(mar), formula.log = art ~ kid5)

Residuals:
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
-2.182360 -1.435045 -0.529781 -0.006112  0.553438 17.305607

Count Coefficients:
                      Estimate  Std.error         Z    P.value
(Intercept)          0.1619557 0.13388600  1.209654 0.22641180
kid5               -10.1873539 0.05368213 -2.001493 0.04533931
phd                  9.6032056 0.03635921  2.521959 0.01167032
factor(mar)Married  21.5084376 0.08488954  2.294906 0.02173851

Zero Coefficients:
               Estimate Std.error         Z      P.value
(Intercept)   -9.143446  1.949244 -4.690765 2.721850e-06
kid5        1002.442278  0.775578  3.094612 1.970704e-03
```

As we can see, we appear to have a -10% chnage in articles published for each kid under 5 a biochemist, 9% increase with each increase in phd prestige index, and a 21% increase if a bio chemist is married

Any further analyses should be done using `emmeans` or other supplementary packages.

4 Insurance Claims

Question: How is the frequency of insurance claims affected by the number of policies taken before hand?

Now, we will use the `insur_claim` (Alomair 2024) dataset, the original dataset has 15 predictors for the model, but we will only use the first 5 for the analysis (since `dat_plot()` maxes out at 15 variables). Below, I load in the dataset and truncate it (Wickham et al. 2019):

```
claim.dat <- read_csv("insur_claim.csv") |>
  select(y,x1,x2,x3,x4,x5)
```

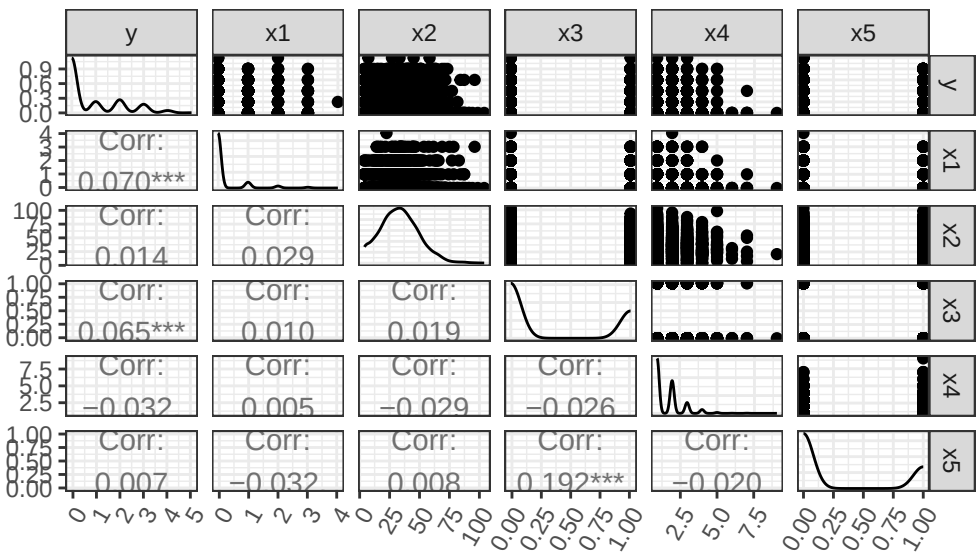
Here is the following predictor annotations list:

- `x1`: Kids to pickup
- `x2`: Time to drive
- `x3`: Car use (0=private, 1 =business)
- `x4`: Number of policies before current policy
- `x5`: Car color (0=non-red, 1=red)

and I use `dat_plot()` to generate a scatter plot matrix:

```
dat_plot(claim.dat)
```

Figure 10: Scatterplot Matrix of Insurance Data



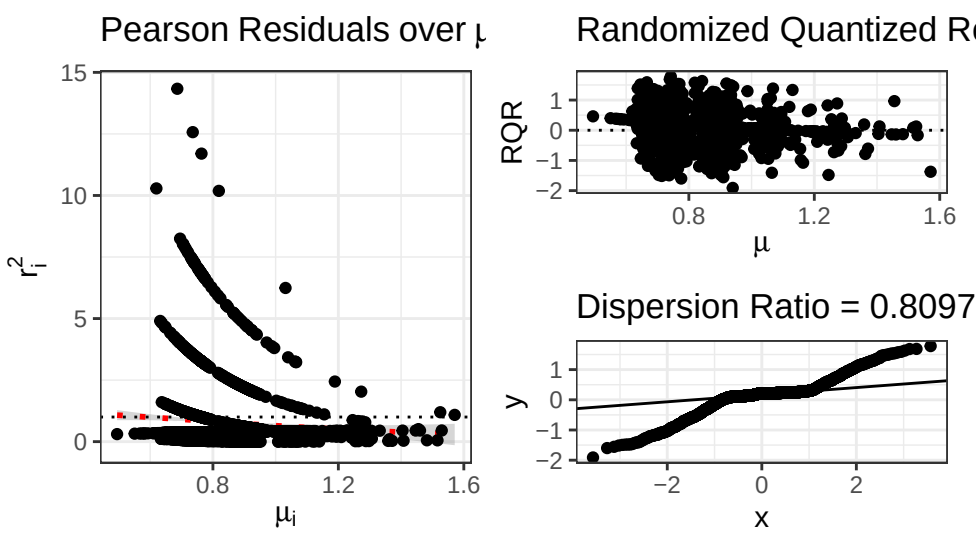
We can't really see much here, we're going to have to use a Generalized Poisson Model (GP2) using the `glm_pois_GP2()` function, we'll also keep the model simple by adding all predictors. Below, I fit a generalized poisson model:

```
mod.claim <- glm_pois_GP2(data=claim.dat, y~x1+x2+factor(x3)+x4+factor(x5))
```

Now, we check the model fit using `resid_diag()`.

```
resid_diag(mod.claim)
```

Figure 11: Residual Diagnostics of Insurance Model.



While we do have an exponential pattern with the residual (implying that we might need to fit a gamma regression model), we will still proceed with the analysis as if the residuals were in order.

Below, I use `interpert()` to print a summary of the model.

```
interpret(mod.claim)
```

```
Call:
glm_pois_GP2(data = claim.dat, formula = y ~ x1 + x2 + factor(x3) +
    x4 + factor(x5))

Residuals:
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
-1.529626 -0.777547 -0.717786 -0.000344  1.060313  4.313614

Count Coefficients:
              Estimate   Std.error  Crit.stat     P.value
(Intercept) -0.2572245 0.096667132 -2.6609305 0.007792505
x1          17.5598150 0.053875705  3.0027836 0.002675226
x2           0.1105198 0.001922616  0.5745230 0.565613929
factor(x3)1 21.8907866 0.064849471  3.0525348 0.002269174
x4          -4.8632592 0.033558749 -1.4856022 0.137384369
factor(x5)1 -2.1745763 0.068563045 -0.3206638 0.748465176
```

Based on the model summary provided, the number of filed insurance claims seems to go down -4.86% for every additional policy that was taken out before the current policy!

Any further analyses should be done using `emmeans` or other supplementary packages.

References

Alomair, Gadir. 2024. "Predictive Performance of Count Regression Models Versus Machine Learning Techniques: A Comparative Analysis Using an Automobile Insurance Claims Frequency Dataset." *PLOS ONE* 19 (12): e0314975. https://doi.org/10.1371/journal.pone.0314975.

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Long, J. Scott. 1990. "The Origins of Sex Differences in Science." *Social Forces* 68 (4): 1297–1316.

McMillan, Garnett Pierre. 2001. *Ache Residential Grouping and Social Foraging.* The University of New Mexico.

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.